



Enhancement of Collection Efficiency of an Inertial Impactor Using an Additional Punched Impaction Plate

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ABSTRACT

In this study, the collection efficiency of a round-nozzle inertial impactor was enhanced through use of an additional punched impaction plate. The additional plate was installed between the nozzle outlet and the existing impaction plate. To determine its effect, a simulation and an experiment were conducted. A parametric study was performed on a $PM_{2.5}$ inertial impactor to improve the optimum efficiency based on the hole diameter of the additional plate and its distance from the existing plate. As a result, the cut-off size was reduced by 27% from 2.52 μm to 1.84 μm . The effect of the additional punched impaction plate on PM_1 and PM_{10} inertial impactors was also investigated. The improvement in efficiency could be observed even when the additional punched impaction plate had varying Reynolds numbers and nozzle sizes. Accordingly, the $(Stk_{50})^{0.5}$ value was reduced by 25% from 0.49 to 0.37 by adding a punched impaction plate in a round-nozzle inertial impactor.

Keywords: Inertial impactor; Collection efficiency; Stokes number; Additional punched impaction plate.

INTRODUCTION

Fine dust and ultrafine dust can have a great impact on our lives and health. In this connection, there is a study that particulate matter is associated with heart and lung function and shortens the life span (Koenig *et al.*, 2005). Furthermore, ultrafine dusts of small size can be deposited in the alveoli and bronchi, and are more dangerous because they can enter the bloodstream through the lung tissue (Schwartz *et al.*, 1996; Harrison and Yin, 2000). Therefore, there is a need to sample and analyse small particles that are fatal to the human body. In order to capture and analyse small particles, it is necessary to develop a high-efficiency aerosol sampler. Sampling of aerosol particles are often followed by chemical analyses, such as chromatography, atomic absorption spectrometry, and particle-induced X-ray emission (Karlsson *et al.*, 1997). Some chemical analysis methods require pre-treatment of samples, which makes it difficult to apply the pre-treated sample to other analyses. Therefore, there is also a need to prepare multiple samples of the same aerosol.

Owing to their simple operation and principles, inertial impactors are widely used to collect particles according to size. The aerosol entering the inlet of an inertial impactor

will go through the nozzle. After being accelerated, the aerosol direction is changed by 90° using an impaction plate. Particles with high inertia cannot follow the sharp bend in the flow and thus are separated from the streamline and collected on the impaction plate. On the other hand, particles with less inertia can easily follow the streamline and escape through the impactor outlet. The particle size corresponding to 50% collection efficiency is known as the cut-off size. By adjusting the cut-off size, particles of a certain size or larger can be selectively collected. To collect particles with smaller sizes, the cut-off size needs to be reduced. Thus, an efficient inertial impactor is required.

Many studies have been conducted to enhance the collection efficiency of inertial impactors. The effects of nozzle diameter, length of nozzle throat, distance between nozzle outlet and impaction plate, Reynolds number, and particle density on collection efficiency have been reported (Marple and Liu, 1974; Rader and Marple, 1985; Grinshpun *et al.*, 2005). A study on the effect of impaction plate diameter and particle density in a round-nozzle inertial impactor has been performed (Huang and Tsai, 2002). A change in cut-off size has been observed according to the distance between the nozzle and impaction plate of an inertial impactor at low pressure (Arffman *et al.*, 2011). Charging the particles in a cascade impactor using electrophoresis can improve collection efficiency (Vinchurkar *et al.*, 2009). Some studies have attempted to capture smaller particles by installing an inertial filter at the end of the impactor (Furuuchi *et al.*, 2010; Hata *et al.*, 2012; Zhang *et al.*, 2016). Meanwhile, the geometry of the inlet has been varied to improve

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collection efficiency. Use of a horizontal inlet instead of the conventional vertical inlet allows particles to be concentrated in the central axis, resulting in a smaller cut-off size (Kim *et al.*, 2014ab). The geometry and condition of the impaction plate can also affect collection efficiency. The thermophoresis effect has been adopted to increase collection efficiency by lowering the temperature of the impaction plate (Lee and Kim, 2002, 2003). Changes in collection efficiency have been investigated using a porous metal substrate on the impaction plate in accordance with various Reynolds numbers and nozzle diameters (Huang *et al.*, 2001). The surface of an impaction plate was coated with grease and Teflon to determine its effect on collection efficiency (Lee *et al.*, 2005). Furthermore, a study has been conducted by varying impaction plate geometry to improve collection efficiency (Tsai and Cheng, 1995). The cut-off size of an inertial impactor was reduced by using an impaction plate with a concave elliptical surface to decrease particle bounce (Kim and Yook, 2011; Kim *et al.*, 2013). In order to capture smaller particles, a new stage was added between the last stage of the ELPI and the filter (Yli-Ojanperä *et al.*, 2010).

Various methods have been studied to improve the collection efficiency of inertial impactors, particularly by varying the geometry of the impaction plate. However, there might be practical restrictions on the actual application of this due to difficulties in replacing the built-in impaction plate of inertial impactors. Thus, the purpose of this study is to improve impactor collection efficiency and prepare multiple samples of the same aerosol by applying an additional punched impaction plate without changing the existing geometry of the inertial impactor.

NUMERICAL METHOD

A sectional view of the round-nozzle inertial impactor with an additional punched impaction plate is shown in Fig. 1. W represents the round nozzle diameter, T is the length of the nozzle throat, S is the distance between the nozzle outlet and the impaction plate, D is the diameter of the impaction plate, W_1 is the diameter of the hole in the additional punched impaction plate, and S_1 is the distance between the original and additional impaction plates.

The commercial ANSYS FLUENT *Release 16.1* code was utilized to simulate the flow field in the round-nozzle inertial impactor. For numerical analysis, the model parameters were set at $W = 0.9$ mm, $S = 3.8$ mm, $T = 1$ mm, and $D = 6$ mm for the PM₁ inertial impactor; $W = 1.6$ mm, $S = 5.4$ mm, $T = 1.6$ mm, and $D = 6$ mm for the PM_{2.5} reference model; and $W = 6.2$ mm, $S = 20.9$ mm, $T = 6.2$ mm, and $D = 20.2$ mm for the PM₁₀ reference model. The parametric study was performed by varying the W_1/W ratio in the range of 8/16–14/16 and the S_1/S ratio in the range of 0.02–0.25 to determine the effect of an additional punched impaction plate.

The flow was assumed to be steady, two-dimensional axisymmetric, and incompressible. Moreover, the laminar flow was assumed in all cases to satisfy a Reynolds number below 3000. The temperature was 20°C and the applied

pressure was 101.3 kPa. For Reynolds numbers below 1500, the effect of gravity was considered in all analyses since it significantly affects collection efficiency (Huang and Tsai, 2001). As can be seen in Fig. 1(b), the velocity inlet condition at the inlet, pressure outlet condition at the outlet, axis condition at the centerline of the impactor, and no-slip condition in all walls were set as the boundary conditions. The SIMPLE scheme for pressure-velocity coupling was used. Continuity, momentum, and energy equations were solved repeatedly by adjusting the convergence criterion at 10^{-6} . The grid dependence test was performed on the PM₁, PM_{2.5}, and PM₁₀ inertial impactors. As a result, about 20000–150000 grid cells were used.

After flow analysis, the behavior of particles was investigated. For the particle behavior analysis, the discrete phase model (DPM) included in the ANSYS FLUENT code was used. Gravitational force, Brownian force, and Stokes drag force with slip correction factor were considered (ANSYS FLUENT User's Guide 15.0, 2013, chap. 24). A particle was assumed to be permanently collected when it hit any wall. The particle density was set at 1000 kg m^{-3} to express the aerodynamic size. To simulate the collection efficiency (η) of the inertial impactor, 500 particles with uniform size and density were arranged at regular intervals from the center of the nozzle inlet to the edge, as shown in Fig. 1(b). Based on simulation results with or without particle collection, the statistical Lagrangian particle-tracking (SLPT) model was used to calculate collection efficiency, as presented in Eq. (1) (Kim and Yook, 2011; Kim *et al.*, 2014b; Lee *et al.*, 2014; Park *et al.*, 2015).

$$\eta = \frac{2\Delta r}{R_{in}^2} \sum_{i=1}^{N_{in}} r_i f_i \quad (1)$$

Here, $\Delta r = R_{in}/(N_{in} - 1)$ is the distance between particles, R_{in} is the inlet radius, N_{in} is the number of particles arranged in the radial direction, $r_i = (i - 1)\Delta r$ is the particle position in the radial direction from the center, i is the particle index and f_i is the particle collection index. In other words, $f_i = 1$ when the i th particle is collected and $f_i = 0$ when the i th particle is not collected.

EXPERIMENTAL METHOD

Inertial impactors were manufactured with cut-off sizes of 1 μm and 2.5 μm . Additional punched impaction plates were fabricated for PM₁ and PM_{2.5} inertial impactors. Fig. 2 shows the experimental scheme of the collection efficiency measurement for the conventional inertial impactor and the impactor installed with an additional punched impaction plate. Polystyrene latex (PSL) spheres were used as the particle in this experiment. The PSL particle has a density of 1050 kg m^{-3} and monodisperse sizes of 0.6, 0.7, 0.8, 0.9, 1.0, 1.6, 2.0, 2.5, 3.0, and 4.0 μm . Both the existing impaction plate and the additional punched impaction plate were coated with vacuum grease. Compressed clean air was injected into the atomizer and the PSL solution contained within the atomizer was sprayed. Subsequently, the moisture

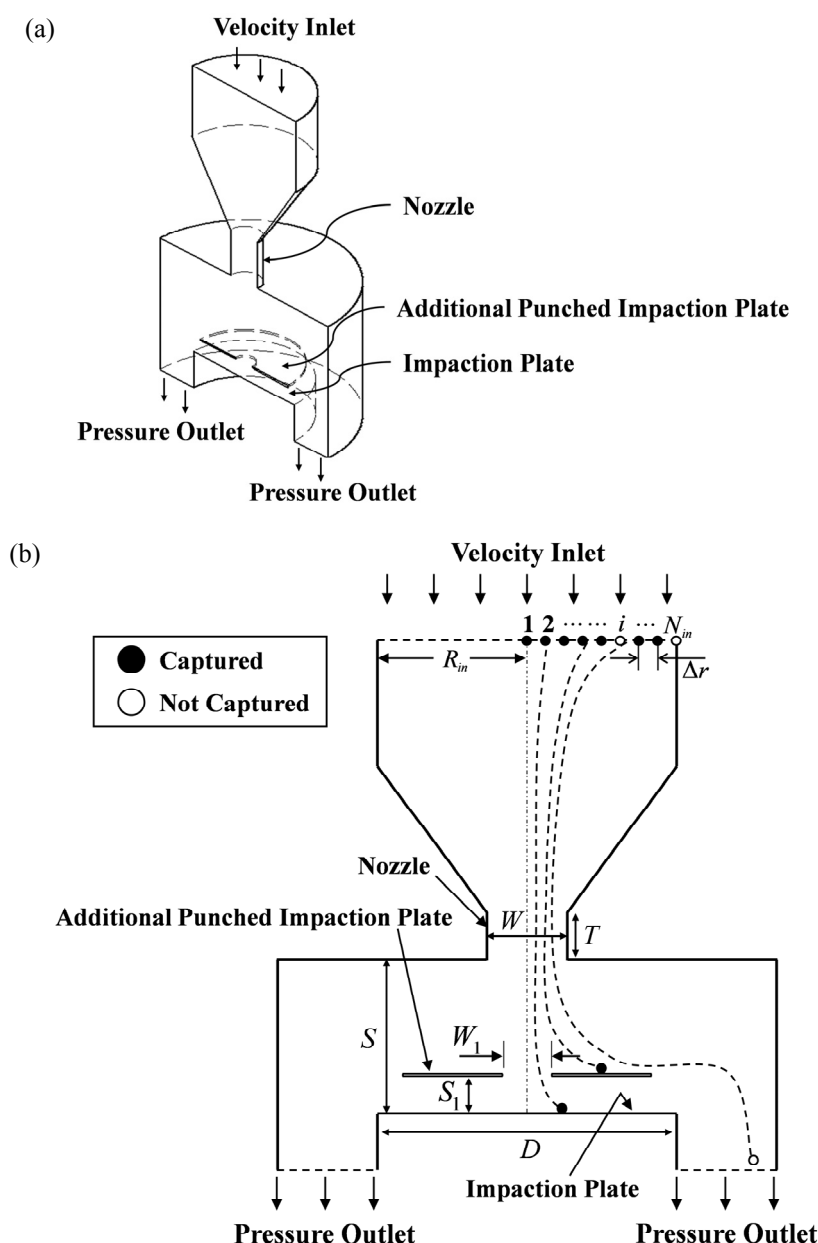


Fig. 1. Schematic of an inertial impactor with an additional punched impaction plate: (a) three-dimensional sectional view; (b) cross-sectional view.

of the generated aerosol particles was removed using a diffusion dryer. To adjust the aerosol flow rate for each impactor, flow rate was measured using a laminar flowmeter and the remaining aerosols were exhausted through the needle valve and HEPA filter. The aerosol was guided into an optical particle counter (OPC) using two three-way valves and the concentration of aerosol was determined using the OPC to measure the upstream (C_{up}) and downstream (C_{down}) particle number concentrations. Through this experimental process, the collection efficiency (η) of the inertial impactor was calculated as follows.

$$\eta = \frac{C_{up} - C_{down}}{C_{up}} \times 100 (\%) \quad (2)$$

The Stokes number (Stk) was calculated as follows.

$$\text{Stk} = \frac{\rho_p d_p^2 C_c U}{9\mu W} \quad (3)$$

Here, ρ_p is the particle density, d_p is the particle size, C_c is the slip correction factor, U is the average jet velocity in the nozzle, μ is the viscosity of air, and W is the diameter of the nozzle. Since the condition of $(\text{Stk}_{50})^{0.5} = 0.49$ should be satisfied when designing a round-nozzle impactor (Hinds, 1999), the aerosol flow rate (Q_{in}) flowing into the impactor and the resulting average velocity in the nozzle (U) were set at $Q_{in} = 2.0$ LPM and $U = 35.8 \text{ m s}^{-1}$ for the PM₁ impactor and $Q_{in} = 2.8$ LPM, and $U = 12.5 \text{ m s}^{-1}$ for

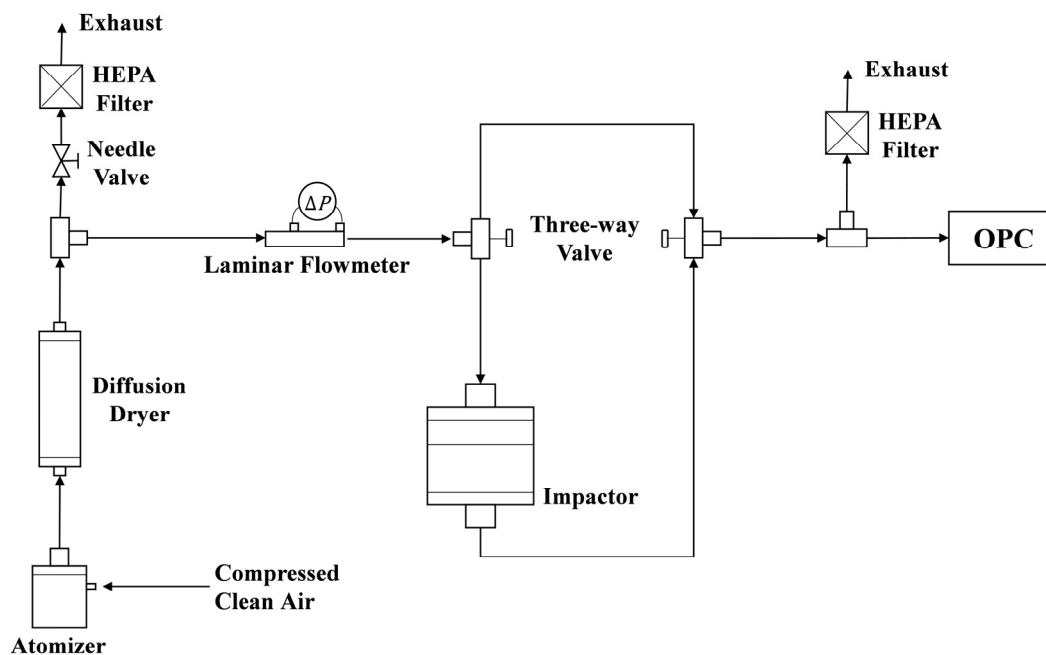


Fig. 2. Schematic of experimental setup.

the $PM_{2.5}$ impactor. At this time, the Reynolds numbers were 2596 and 1807, respectively.

RESULTS AND DISCUSSION

Fig. 3 shows the variation in cut-off size obtained from the numerical parametric study based on the hole diameter (W_1) of the additional punched impaction plate and its distance from the existing impaction plate (S_1) in the reference $PM_{2.5}$ impactor ($W = 1.6$ mm, $S = 5.4$ mm, $T = 1.6$ mm, $D = 6$ mm). The lowest cut-off size of 1.95 μ m was achieved when the ratio of the additional impaction plate diameter to the nozzle diameter was equal to $W_1/W = 10/16$, as can be seen in Fig. 3(a). On the other hand, the cut-off size was minimized to 1.84 μ m when the distance between the existing and additional impaction plates was $S_1/S = 0.1$, as presented in Fig. 3(b). However, for smaller impactors such as PM_1 impactors, it was difficult to manufacture an impactor with $S_1/S = 0.1$. Instead, the S_1/S ratio was set at 0.15 by considering the manufacturability. As a result, the parameters of the PM_1 impactor designed for the experiment were $W = 1.1$ mm, $T = 1.2$ mm, $S = 4.6$ mm, $D = 9$ mm, $W_1 = 0.7$ mm, and $S_1 = 0.7$ mm. The parameters of the $PM_{2.5}$ impactor were $W = 2.2$ mm, $T = 2.2$ mm, $S = 7.4$ mm, $D = 9$ mm, $W_1 = 1.4$ mm, $S_1 = 1.1$ mm. The additional punched impaction plate in the prepared PM_1 and $PM_{2.5}$ inertial impactors had a thickness of 2 mm and an outer diameter of 8 mm.

The collection efficiency of the prepared impactors obtained from the experiment and simulation were compared and are shown in Fig. 4. Accordingly, the accuracy of the simulation method used in this study was confirmed. From the PM_1 impactor results in Fig. 4(a), the cut-off size was found to be 1.0 μ m if only the existing impaction plate was applied. On the other hand, the cut-off size was reduced to

0.74 μ m by installation of an additional punched impaction plate. Likewise, the cut-off size of the $PM_{2.5}$ impactor was reduced from 2.56 μ m to 1.94 μ m through installation of an additional punched impaction plate, as shown in Fig. 4(b). Since it was impossible to practically aerosolize PSL particles larger than 4 μ m using an atomizer in the PM_{10} impactor, the trend in collection efficiency for PM_{10} was observed only by simulation. As a result, the cut-off size was estimated to decrease from 9.89 μ m to 7.61 μ m with the installation of an additional impaction plate. Thus, the additional impaction plate was confirmed to improve collection efficiency by reducing the cut-off size of the inertial impactor.

To investigate the cut-off size reduction mechanism of an inertial impactor with an additional punched impaction plate, the flow and particle trajectory in the impactor were simulated and observed. The flow streamline with and without the additional punched impaction plate are compared in Fig. 5(a) and the particle trajectory injected at the same position is compared in Fig. 5(b). In the case of the inertial impactor without an additional impaction plate, the flow hit the impaction plate and changed direction by 90° after being accelerated through the nozzle, as shown in the left-hand-side figure of Fig. 5(a). In this process, the particles entering from the center of the inlet were collected by the impaction plate. The particles entering from the edge of the inlet could not be collected, but were instead released from the impactor since the stream entering the edge of the impactor inlet passed through the farthest height from the impaction plate, as can be seen in the left-hand-side figure of Fig. 5(b). In the case of the inertial impactor with an additional punched impaction plate, the flow was divided into two parts, as shown in the right-hand-side figure of Fig. 5(a). The flow entering at the center of the inlet passed through the hole of the additional impaction plate and then over the existing impaction plate, while the flow entering

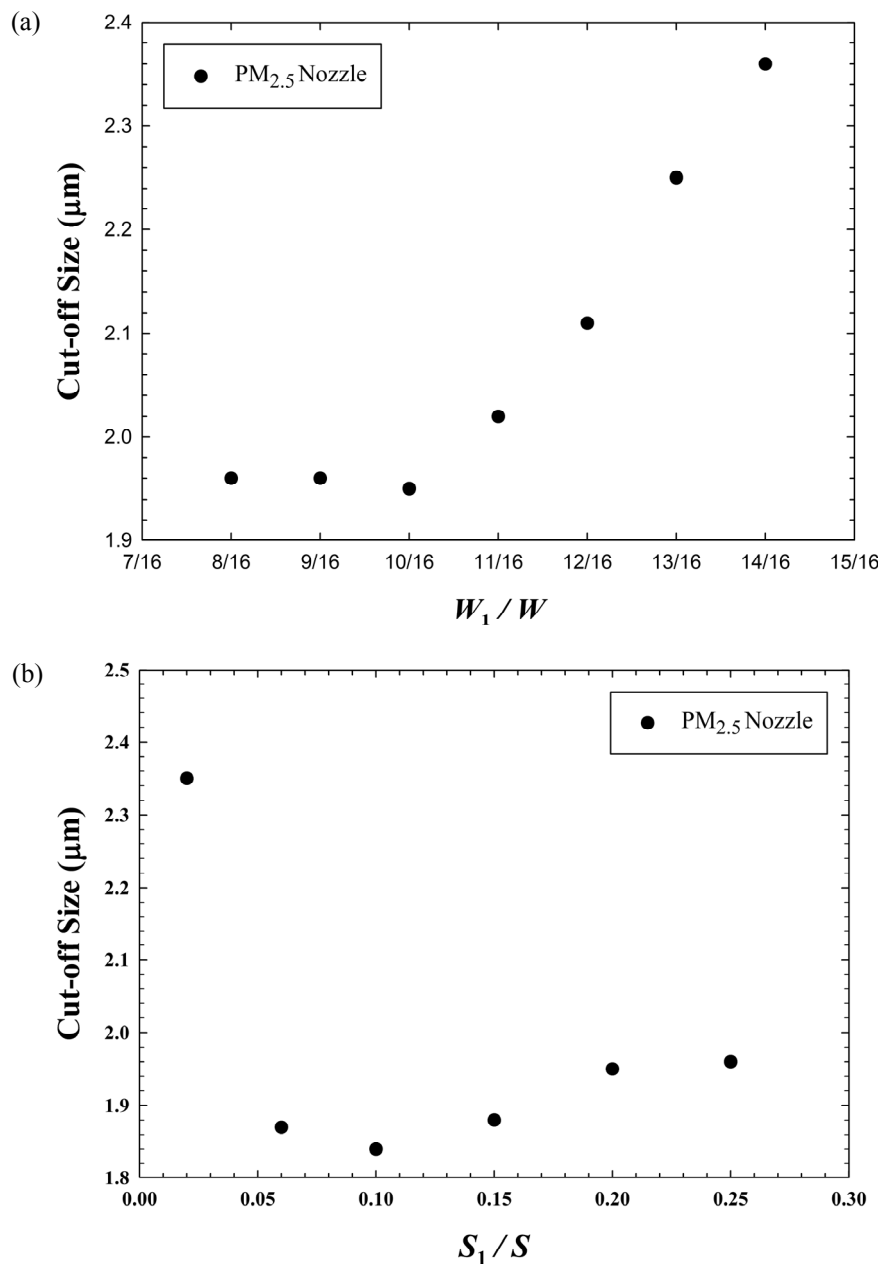


Fig. 3. Variation in cut-off size of the reference $\text{PM}_{2.5}$ impactor: (a) effect of the hole diameter; (b) effect of the distance between the existing impactation plate and the additional punched impactation plate.

from the edge of the inlet passed over the additional impactation plate. Due to these flows, the particles entering in the vicinity of the inlet center were collected at the existing impactation plate, while the particles entering the edge of the inlet were collected at the additional impactation plate, as shown in the right-hand-side figure of Fig. 5(b). Hence, the overall collection efficiency could be improved by effectively dividing the stream from the center and the edge of the inlet since the particles from the edge of the inlet could be collected on the additional impactation plate.

In addition, the effect of the additional impactation plate on pressure drop across the impactor was examined. In case of the PM_1 impactor, the pressure drops with and without the additional impactation plate were determined to be 1080

Pa and 1097 Pa, respectively, from numerical simulation, and those were 1155 Pa and 1157 Pa, respectively, from experimental measurement. For the $\text{PM}_{2.5}$ impactor, the pressure drops with and without the additional impactation plate were numerically obtained to be 132 Pa and 135 Pa, respectively, and those were experimentally determined to be 140 Pa and 142 Pa, respectively. As a result, no significant difference in pressure drop was found between the cases with and without the additional impactation plate.

Fig. 6 shows the simulation results of collection efficiency as a function of nozzle diameter and Reynolds number with or without the additional punched impactation plate. The square-root of the Stokes number is indicated in the x-axis of the graph. The nozzle size, aerosol flow rate, and the

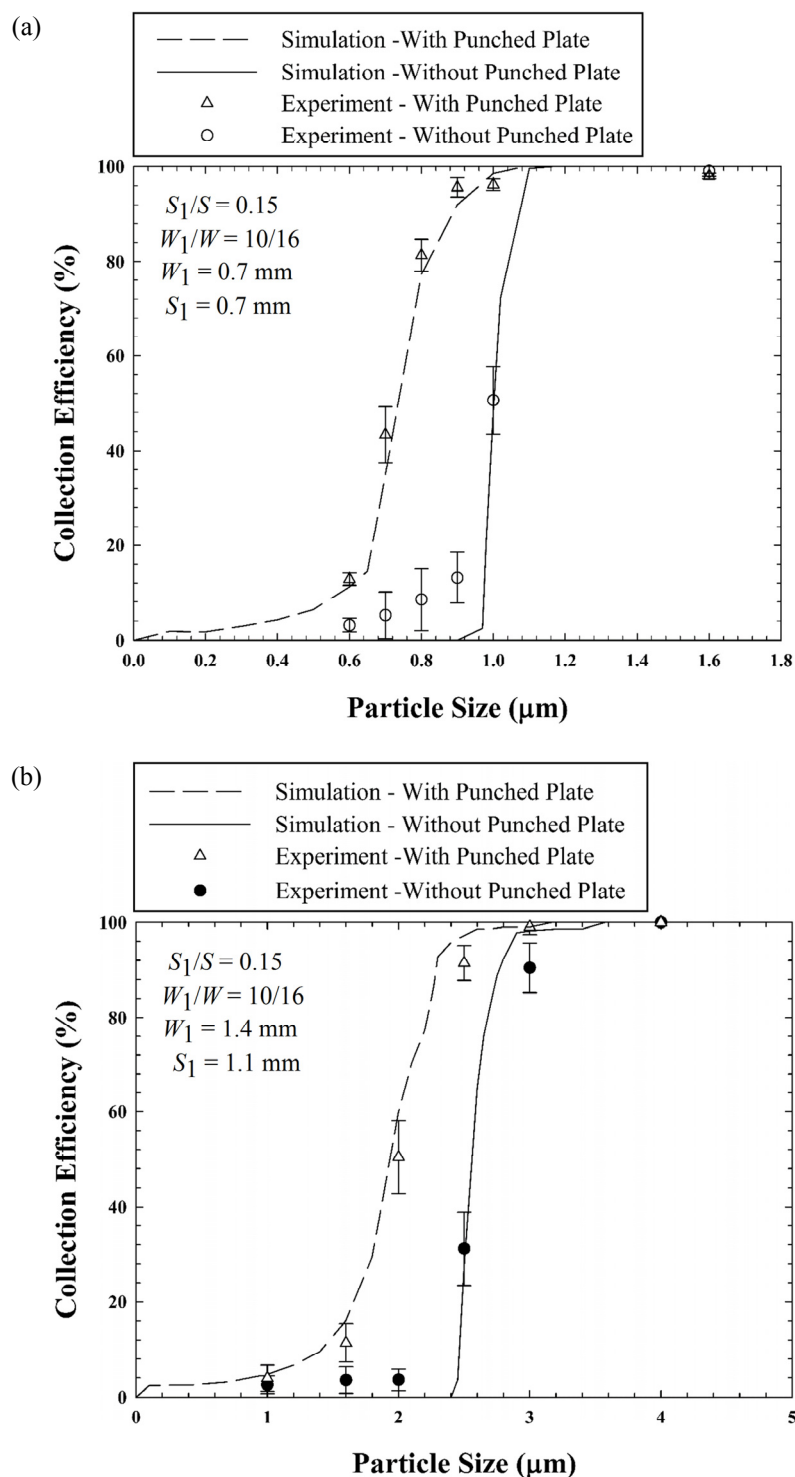


Fig. 4. Comparison of collection efficiency between numerical results and experimental data: (a) PM_1 impactor; (b) $PM_{2.5}$ impactor.

corresponding nozzle Reynolds number of various inertial impactor models used in the numerical parametric study are presented in Table 1. At this time, the Reynolds number was varied within the range of about 500 to 2000. The results indicate a specific tendency in the collection efficiency data as a function of Stokes number with or without the additional impactation plate. Thus, the fitting curve was

derived for each case with or without the additional punched impactation plate. From the fitting curve comparison, $(Stk_{50})^{0.5} = 0.49$ when using only the existing impactation plate, while $(Stk_{50})^{0.5} = 0.37$ when the additional punched impactation plate was installed, showing a decrease of 25% compared to 0.49. In addition, the sharpness of the collection efficiency curve, that is, the ratio of the particle size corresponding to

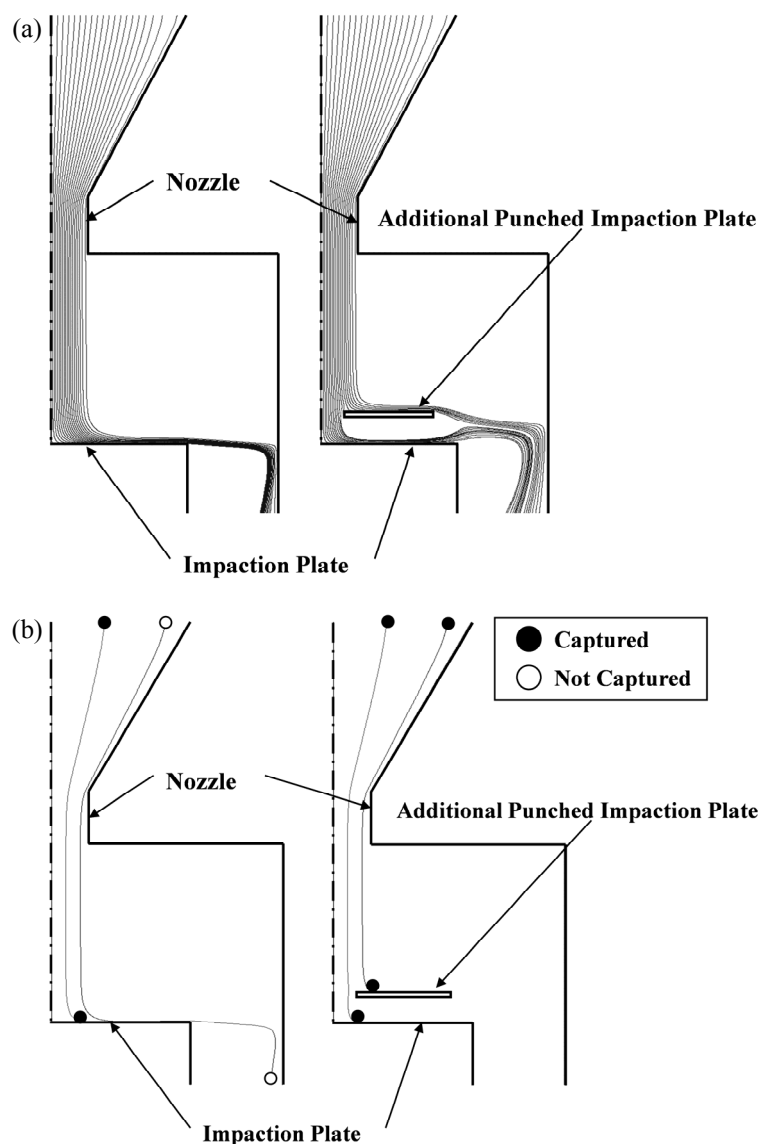


Fig. 5. Comparison between inertial impactors with and without the additional punched impaction plate: (a) streamline; (b) particle trajectory.

84% collection efficiency to the particle size corresponding to 16% collection efficiency (Kavouras and Koutrakis, 2001; Misra *et al.*, 2002), was estimated to be 1.073 without the additional punched impaction plate, while the sharpness was 1.225 with the additional punched impaction plate, showing an increase of 14%.

CONCLUSIONS

In this study, an additional punched impaction plate was employed to increase the collection efficiency of a round-nozzle inertial impactor. The hole diameter of the additional plate and the distance from the existing impaction plate were selected as the geometric parameters of the former, and the cut-off size was simulated by varying these parameters. As a result, the ratios of $W_1/W = 10/16$ and $S_1/S = 0.15$ were applied to prepared PM_1 and $PM_{2.5}$ impactors, taking manufacturability into consideration. The collection

efficiency simulation results of the prepared impactor models agreed with the experimental results. The cut-off size of an inertial impactor was found to be smaller with the additional plate than in cases without it, being reduced from 1.0 μm to 0.74 μm for the PM_1 impactor, 2.56 μm to 1.94 μm for the $PM_{2.5}$ impactor, and 9.89 μm to 7.61 μm for the PM_{10} impactor. Several models were established for PM_1 , $PM_{2.5}$, and PM_{10} impactors based on different nozzle sizes and aerosol flow rates using a simulation of the method proven by the experiments. The simulated collection efficiency was obtained by using a nozzle Reynolds number within the range of 500–2000. For the inertial impactors with and without an additional punched impaction plate, specific fitting curves could be observed from the collection efficiency data, resulting in $(Stk_{50})^{0.5} = 0.49$ without the additional impaction plate and $(Stk_{50})^{0.5} = 0.37$ with the additional plate. A decrease of approximately 25% was achieved. In conclusion, the cut-off size of a conventional

inertial impactor can be efficiently reduced, while collection efficiency is improved, simply by adding another punched impaction plate. The installation of this additional plate

can also be useful in preparing multiple samples of the same aerosol.

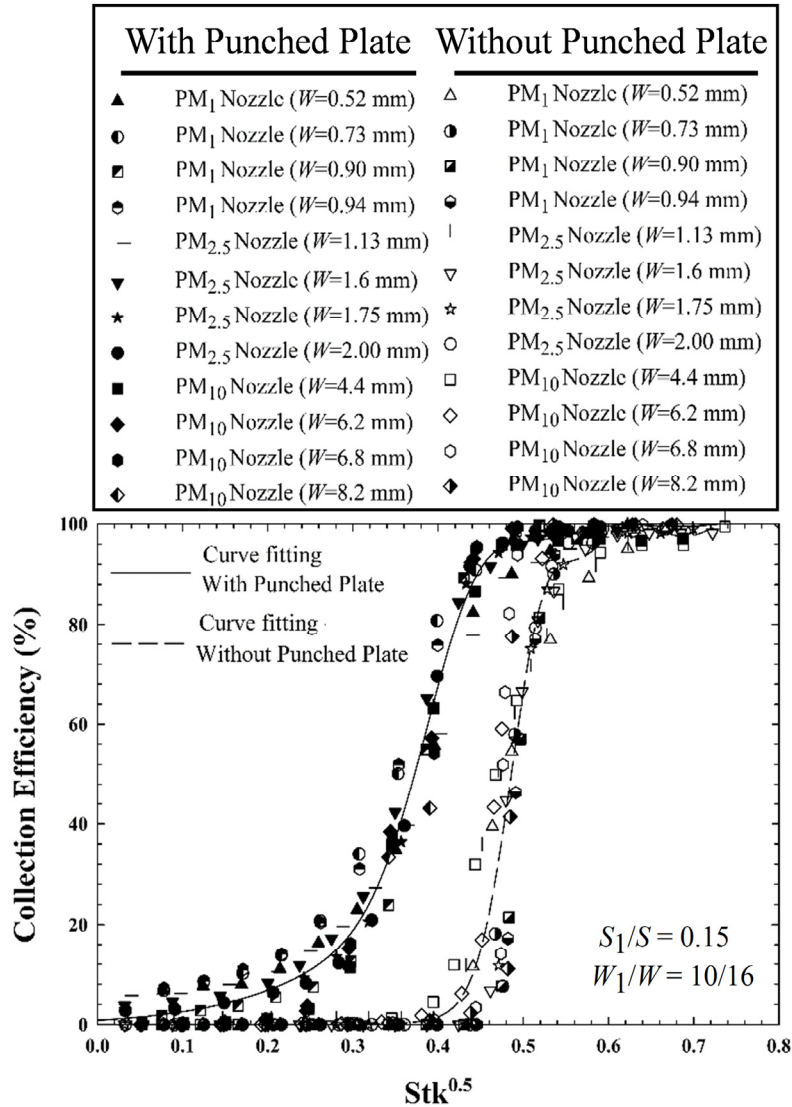


Fig. 6. Comparison of collection efficiency between inertial impactors with and without the additional punched impaction plate.

Table 1. Nozzle sizes and aerosol flow rates assumed for the numerical parametric study.

Inertial Impactor	W [mm]	Q_{in} [L min ⁻¹]	Nozzle Reynolds Number
PM ₁ Nozzle ($S_1/S = 0.15$, $W_1/W = 10/16$)	0.52	0.22	600
	0.73	0.63	1201
	0.90	1.1	1711
	0.94	1.34	2000
PM _{2.5} Nozzle ($S_1/S = 0.15$, $W_1/W = 10/16$)	1.13	0.4	501
	1.6	1.1	962
	1.75	1.5	1200
	2	2.29	1600
PM ₁₀ Nozzle ($S_1/S = 0.15$, $W_1/W = 10/16$)	4.4	1.57	500
	6.2	4.1	926
	6.8	5.83	1200
	8.2	9.96	1700

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